7 **(a)** State what is meant by a *photon*.

.....

.....

.....[2]

(b) Light in a beam has a continuous spectrum that lies within the visible region. The photons of light have energies ranging from 1.60 eV to 2.60 eV.

The beam passes through some hydrogen gas. It then passes through a diffraction grating and an absorption spectrum is observed.

(i) All of the light absorbed by the hydrogen is re-emitted. Explain why dark lines are still observed in the absorption spectrum.

.....

.....[1]

(ii) Some energy levels for the electron in an isolated hydrogen atom are illustrated in Figure 7.1.

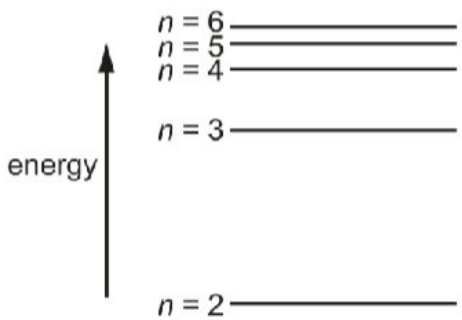


Figure 7.1

Table 7.1 shows the wavelengths of photons that are emitted in the transitions to $n = 2$ from the other energy levels shown in Figure 7.1. The energy associated with the energy level $n = 2$ is –3.40 eV.

Table 7.1

wavelength / nm
412
435
488
658

Calculate the energy, in J, of energy level $n = 3$.

energy = ... J [2]

(iii) Explain why the energy of each energy level is negative.

.....

.....[1]

[Total: 6]